

Tong QFT — Problem Sheet 1

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SO(3) Symmetry, Q4

Verify that the Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi_a \partial^\mu \phi_a - \frac{1}{2} m^2 \phi_a \phi_a \quad (1.1)$$

for a triplet of real fields ϕ_a ($a = 1, 2, 3$) is invariant under the infinitesimal $SO(3)$ rotation by θ

$$\phi_a \rightarrow \phi_a + \theta \epsilon_{abc} n_b \phi_c \quad (1.2)$$

where n_a is a unit vector. Compute the Noether current j^μ . Deduce that the three quantities

$$Q_a = \int d^3x \epsilon_{abc} \dot{\phi}_b \phi_c \quad (1.3)$$

are all conserved and verify this directly using the field equations satisfied by ϕ_a .

Solution. Under the $SO(3)$ rotation, the $\phi\phi$ term transform as

$$\phi_a \phi_a \rightarrow \phi_a \phi_a + 2\theta \epsilon_{abc} \phi_a n_b \phi_c + \mathcal{O}(\theta^2) = \phi_a \phi_a. \quad (1.4)$$

Therefore, the Lagrangian is invariant under the transformation.

The corresponding conserved current is given by

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a = \partial^\mu \phi_a \epsilon_{abc} n_b \phi_c = -n_b J_b^\mu. \quad (1.5)$$

Since the conservation law must hold for arbitrary unit vector, n_a , all three components of the current are conserved separately. Integrate the time component over the space, we find the conserved charge

$$Q_a = \int d^3x J_a^0 = \int d^3x \epsilon_{abc} \dot{\phi}_b \phi_c. \quad (1.6)$$

The equation of motion is given by

$$(\partial^2 - m^2)\phi_a = 0. \quad (1.7)$$

We use this to verify the charge conservation:

$$\frac{dQ_a}{dt} = \int d^3x \epsilon_{abc} (\ddot{\phi}_b \phi_c + \dot{\phi}_b \dot{\phi}_c) = \int d^3x \epsilon_{abc} \phi_c (\nabla^2 + m^2)\phi_b = \int d^3x \epsilon_{abc} (\nabla \phi_b) \cdot (\nabla \phi_c) = 0, \quad (1.8)$$

where we have used the antisymmetric property of ϵ_{abc} consistently.

Infinitesimal Lorentz Transformation, Q5

A Lorentz transformation $x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu$ is such that it preserves the Minkowski metric $\eta_{\mu\nu}$, meaning that $\eta_{\mu\nu} x^\mu x^\nu = \eta_{\mu\nu} x'^\mu x'^\nu$ for all x . Show that this implies that

$$\eta_{\mu\nu} = \eta_{\sigma\tau} \Lambda^\sigma{}_\mu \Lambda^\tau{}_\nu. \quad (2.1)$$

Use this result to show that an infinitesimal transformation of the form

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu \quad (2.2)$$

is a Lorentz transformation when $\omega^{\mu\nu}$ is antisymmetric: i.e. $\omega^{\mu\nu} = -\omega^{\nu\mu}$.

Write down the matrix form for $\omega^\mu{}_\nu$ that corresponds to a rotation through an infinitesimal angle θ about the x^3 -axis. Do the same for a boost along the x^1 -axis by an infinitesimal velocity v .

Solution. Substitute the Lorentz transformation into the metric preserving statement, we find

$$\eta_{\mu\nu} x^\mu x^\nu = \eta_{\sigma\tau} \Lambda^\sigma{}_\mu \Lambda^\tau{}_\nu x^\mu x^\nu \implies \eta_{\mu\nu} = \eta_{\sigma\tau} \Lambda^\sigma{}_\mu \Lambda^\tau{}_\nu. \quad (2.3)$$

For an infinitesimal transformation of the form, it needs to satisfy the above equality

$$\eta_{\mu\nu} = \eta_{\sigma\tau} (\delta^\sigma{}_\mu + \omega^\sigma{}_\mu) (\delta^\tau{}_\nu + \omega^\tau{}_\nu) = \eta_{\mu\nu} + \omega_{\nu\mu} + \omega_{\mu\nu} + \mathcal{O}(\omega^2). \quad (2.4)$$

Therefore, we require $\omega_{\mu\nu} + \omega_{\nu\mu} = 0$, i.e. $\omega^{\mu\nu} = -\omega^{\nu\mu}$.

For a rotation through an infinitesimal angle θ about the x^3 -axis, the coordinates transform as $x^1 \rightarrow x^1 + \theta x^2$ and $x^2 \rightarrow x^2 - \theta x^1$, so $\omega^\mu{}_\nu$ takes the form

$$\omega^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \theta & 0 \\ 0 & -\theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (2.5)$$

Lowering the first index with $\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$ gives

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\theta & 0 \\ 0 & \theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (2.6)$$

which is manifestly antisymmetric.

For a boost along the x^1 -axis by an infinitesimal velocity v , the coordinates transform as $x^0 \rightarrow x^0 - vx^1$ and $x^1 \rightarrow x^1 - vx^0$, so

$$\omega^\mu{}_\nu = \begin{pmatrix} 0 & -v & 0 & 0 \\ -v & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (2.7)$$

Lowering the first index gives

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & -v & 0 & 0 \\ v & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (2.8)$$

which is again antisymmetric, as required.

Lorentz Invariance and the Energy-Momentum Tensor, Q6

Consider the infinitesimal form of the Lorentz transformation derived in the previous question: $x^\mu \rightarrow x^\mu + \omega^\mu{}_\nu x^\nu$. Show that a scalar field transforms as

$$\phi(x) \rightarrow \phi'(x) = \phi(x) - \omega^\mu{}_\nu x^\nu \partial_\mu \phi(x) \quad (3.1)$$

and hence show that the variation of the Lagrangian density is a total derivative

$$\delta \mathcal{L} = -\partial_\mu (\omega^\mu{}_\nu x^\nu \mathcal{L}). \quad (3.2)$$

Using Noether's theorem deduce the existence of the conserved current

$$j^\mu = -\omega^\rho{}_\nu [T^\mu{}_\rho x^\nu] \quad (3.3)$$

The three conserved charges arising from spatial rotational invariance define the *total angular momentum* of the field. Show that these charges are given by,

$$Q_i = \epsilon_{ijk} \int d^3x (x^j T^{0k} - x^k T^{0j}) \quad (3.4)$$

Derive the conserved charges arising from invariance under Lorentz boosts. Show that they imply

$$\frac{d}{dt} \int d^3x (x^i T^{00}) = \text{constant} \quad (3.5)$$

and interpret this equation.

Solution. Under an infinitesimal Lorentz transformation, the scalar field transforms as

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x) = \phi(x^\mu - \omega^\mu{}_\nu x^\nu) = \phi(x) - \omega^\mu{}_\nu x^\nu \partial_\mu \phi(x). \quad (3.6)$$

Since the Lagrangian is a scalar, it must transform in the same way as the free scalar field, that is

$$\delta \mathcal{L} = -\partial_\mu (\omega^\mu{}_\nu x^\nu \mathcal{L}). \quad (3.7)$$

For a Lorentz invariant Lagrangian, the conserved current is given by

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi - F^\mu = -\omega^\rho{}_\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\rho \phi - \delta^\mu_\rho \mathcal{L} \right) x^\nu = -\omega^\rho{}_\nu T^\mu{}_\rho x^\nu \quad \text{with} \quad F^\mu = -\omega^\mu{}_\nu x^\nu \mathcal{L}.$$

By the antisymmetric property of ω , we find

$$\omega_{\rho\nu} \partial_\mu (T^{\mu\rho} x^\nu) = \frac{1}{2} \omega_{\rho\nu} \partial_\mu (T^{\mu\rho} x^\nu - T^{\mu\nu} x^\rho) = 0 \quad (3.9)$$

So, we have the following conservation equation

$$\partial_\mu (T^{\mu\rho} x^\nu - T^{\mu\nu} x^\rho) = 0, \quad (3.10)$$

which gives the conserved charges

$$Q^{\mu\nu} = \int d^3x (T^{0\mu} x^\nu - T^{0\nu} x^\mu). \quad (3.11)$$

The Lorentz group has six independent parameters: three rotations and three boosts. For spatial rotations, we pick $\rho, \nu \in \{1, 2, 3\}$. The three angular momentum charges are

$$Q^{jk} = \int d^3x (T^{0j} x^k - T^{0k} x^j), \quad (3.12)$$

which we can package into a 3-vector using the Levi-Civita symbol,

$$Q_i = \frac{1}{2} \epsilon_{ijk} Q^{jk} = \epsilon_{ijk} \int d^3x (x^j T^{0k} - x^k T^{0j}). \quad (3.13)$$

These are the components of the total angular momentum of the field.

For Lorentz boosts, we pick $\rho = 0$ and $\nu = i \in \{1, 2, 3\}$, giving the three boost charges

$$Q^{0i} = \int d^3x (T^{00} x^i - T^{0i} x^0) = \int d^3x T^{00} x^i - t \int d^3x T^{0i}. \quad (3.14)$$

Conservation of Q^{0i} states that

$$\frac{d}{dt} \int d^3x (T^{00} x^i - T^{0i} x^0) = 0 \implies \frac{d}{dt} \int d^3x x^i T^{00} = \int d^3x T^{0i} = P^i, \quad (3.15)$$

where we used conservation of momentum $\dot{P}^i = 0$, by translational invariance. Since P^i is constant, this gives

$$\frac{d}{dt} \int d^3x (x^i T^{00}) = \text{constant}. \quad (3.16)$$

This is the field-theoretic analogue of the statement that the centre of energy moves with constant velocity P^i/E , i.e. the centre of energy travels in a straight line at constant speed, in direct analogy with Newton's first law for the centre of mass.

Energy-Momentum Tensor of the Electromagnetic Field, Q7

Maxwell's Lagrangian for the electromagnetic field is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (4.1)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and A_μ is the 4-vector potential. Show that $\mathcal{L}$ is invariant under gauge transformations

$$A_\mu \rightarrow A_\mu + \partial_\mu \xi \quad (4.2)$$

where $\xi = \xi(x)$ is a scalar field with arbitrary (differentiable) dependence on x .

Use Noether's theorem, and the spacetime translational invariance of the action, to construct the energy-momentum tensor $T^{\mu\nu}$ for the electromagnetic field. Show that the resulting object is neither symmetric nor gauge invariant. Consider a new tensor given by

$$\Theta^{\mu\nu} = T^{\mu\nu} - F^{\rho\mu} \partial_\rho A^\nu \quad (4.3)$$

Show that this object also defines four conserved currents. Moreover, show that it is symmetric, gauge invariant and traceless.

Comment: $T^{\mu\nu}$ and $\Theta^{\mu\nu}$ are both equally good definitions of the energy-momentum tensor. However $\Theta^{\mu\nu}$ clearly has the nicer properties. Moreover, if you couple Maxwell's Lagrangian to general relativity then it is $\Theta^{\mu\nu}$ which appears in Einstein's equations.

Solution. It is trivial to see that the Lagrangian is invariant under gauge transformation.

We now look at the conjugate momentum of the Lagrangian

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{2} (\delta_\alpha^\mu \delta_\beta^\nu - \delta_\alpha^\nu \delta_\beta^\mu) (\partial^\alpha A^\beta - \partial^\beta A^\alpha) = -\frac{1}{2} F^{\mu\nu} + \frac{1}{2} F^{\nu\mu} = -F^{\mu\nu}. \quad (4.4)$$

Now consider the translation, $x^\mu \rightarrow x^\mu + \epsilon^\mu$. The potential transform as

$$A^\mu \rightarrow A^\mu - \epsilon^\nu \partial_\nu A^\mu. \quad (4.5)$$

The Lagrangian transform as

$$\mathcal{L} \rightarrow \mathcal{L} - \epsilon^\mu \partial_\mu \mathcal{L}. \quad (4.6)$$

Now, we find

$$\delta \mathcal{L} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \delta A_\nu \right) \implies -\partial_\nu \mathcal{L} = \partial_\mu (F^{\mu\rho} \partial^\nu A_\rho), \quad (4.7)$$

which gives the energy-momentum tensor

$$T^{\mu\nu} = -F^{\mu\rho} \partial^\nu A_\rho - \eta^{\mu\nu} \mathcal{L}. \quad (4.8)$$

It is clear that F and $\mathcal{L}$ are invariant under gauge transformation. However, $\partial_\mu A_\nu$ itself is not invariant under the gauge transformation, so that the energy-momentum tensor is not invariant. It is also clear to see that the first term is not symmetric.

Defining the new object

$$\boxed{\Theta^{\mu\nu} = T^{\mu\nu} - F^{\rho\mu} \partial_\rho A^\nu = -F^{\mu\rho} \partial^\nu A_\rho - F^{\rho\mu} \partial_\rho A^\nu - \eta^{\mu\nu} \mathcal{L} = F^{\rho\mu} F^\nu{}_\rho - \eta^{\mu\nu} \mathcal{L}}, \quad (4.9)$$

which is clearly gauge invariant and symmetric. To see that the above quantity is also a conserved current, consider

$$\partial_\mu (F^{\mu\rho} \partial_\rho A_\nu) = \partial_\mu F^{\mu\rho} \partial_\rho A_\nu + F^{\mu\rho} \partial_\mu \partial_\rho A_\nu = 0, \quad (4.10)$$

where the first term vanishes by the equation of motion and the second term vanishes by the anti-symmetric property of $F^{\mu\nu}$. Therefore,

$$\partial_\mu \Theta^{\mu\nu} = 0. \quad (4.11)$$

Now, consider the trace of the object, which is

$$\Theta^\mu{}_\mu = \eta_{\mu\nu} \Theta^{\mu\nu} = F^{\mu\rho} F_{\mu\rho} - \frac{1}{4} \delta^\mu{}_\mu F_{\rho\sigma} F^{\rho\sigma} = F^{\mu\rho} F_{\mu\rho} - F_{\rho\sigma} F^{\rho\sigma} = 0. \quad (4.12)$$

The Proca Field, Q8

The Lagrangian density for a massive vector field C_μ is given by

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 C_\mu C^\mu \quad (5.1)$$

where $F_{\mu\nu} = \partial_\mu C_\nu - \partial_\nu C_\mu$. Derive the equations of motion and show that when $m \neq 0$ they imply

$$\partial_\mu C^\mu = 0. \quad (5.2)$$

Further show that C_0 can be eliminated completely in terms of the other fields by

$$\partial_i \partial^i C_0 + m^2 C_0 = \partial^i \dot{C}_i. \quad (5.3)$$

Construct the canonical momenta Π_i conjugate to C_i , $i = 1, 2, 3$ and show that the canonical momentum conjugate to C_0 is vanishing. Construct the Hamiltonian density $\mathcal{H}$ in terms of C_0 , C_i and Π_i . (Note: Do not be concerned that the canonical momentum for C_0 is vanishing. C_0 is non-dynamical --- it is determined entirely in terms of the other fields using equation (5.3)).

Solution. The equation of motion trivially follows from (4.4), that is

$$\partial_\mu F^{\mu\nu} + m^2 C^\nu = 0. \quad (5.4)$$

For non-zero mass, take ∂_ν of the above and use the antisymmetric property of $F^{\mu\nu}$, we find

$$\partial_\mu C^\mu = 0. \quad (5.5)$$

Therefore, we can write the following two equations

$$\partial_0 C^0 = \partial_i C^i, \quad (5.6)$$

$$\partial_\mu \partial^\mu C_0 - \partial_0 \partial_\mu C^\mu + m^2 C_0 = \partial_\mu \partial^\mu C_0 + m^2 C_0 = 0. \quad (5.7)$$

Together, we find

$$\partial_i \partial^i C_0 + m^2 C_0 = \partial^i \dot{C}_i. \quad (5.8)$$

The conjugate momenta are given by

$$\Pi_\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} = -F^{0\mu}. \quad (5.9)$$

For $\mu = 0$, the antisymmetric tensor vanishes, and thus $\Pi_0 = 0$. For $\mu = i$, we have $\Pi_i = F_{0i}$. The Hamiltonian density is given by

$$\mathcal{H} = \Pi_\mu \dot{C}^\mu - \mathcal{L}. \quad (5.10)$$

Let's write the Lagrangian in terms of the conjugate momenta:

$$\mathcal{L} = \frac{1}{2} \Pi_i^2 - \frac{1}{4} F_{ij}^2 + \frac{1}{2} m^2 (C_0^2 - C_i^2). \quad (5.11)$$

The Hamiltonian density is

$$\mathcal{H} = \Pi_i^2 + \Pi_i \partial_i C_0 - \mathcal{L} = \frac{1}{2} \Pi_i^2 + \frac{1}{4} F_{ij}^2 - \frac{1}{2} m^2 (C_0^2 - C_i^2) + \Pi_i \partial_i C_0. \quad (5.12)$$

Scale Invariance and the Dilatation Current, Q9

A class of interesting theories are invariant under the scaling of all lengths by

$$x^\mu \rightarrow (x')^\mu = \lambda x^\mu \quad \text{and} \quad \phi(x) \rightarrow \phi'(x) = \lambda^{-D} \phi(\lambda^{-1} x) \quad (6.1)$$

Here D is called the *scaling dimension* of the field. Consider the action for a real scalar field given by

$$S = \int d^4x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - g \phi^p \right) \quad (6.2)$$

Find the scaling dimension D such that the derivative terms remain invariant. For what values of m and p is the scaling (23) a symmetry of the theory. How do these conclusions change for a scalar field living in an $(n + 1)$ -dimensional spacetime instead of a $3 + 1$ -dimensional spacetime?

In $3 + 1$ dimensions, use Noether's theorem to construct the conserved current D^μ associated to scaling invariance.

Solution. The derivative term transform as $\partial_\mu \rightarrow \lambda^{-1} \partial_\mu$, and the integral measure transforms as $d^4x \rightarrow \lambda^4 d^4x$. For the dynamic term to be invariant, we need $\lambda^{4-2-2D} = 1$, which means $D = 1$. For the mass term to be invariant, we require $m^2 \lambda^{4-2D} = 1$, which implies $m = \lambda^{-2}$. For the interaction term, we need $\lambda^{4-pD} = 1$, which gives $p = 4$. For an $(n + 1)$ -dimensional world, we would have $D = (n - 1)/2$, $m = \lambda^{-2}$, and $p = 2(n + 1)/(n - 1)$.

Consider the infinitesimal scaling, $\lambda = 1 + \epsilon$. To first order of ϵ , we find

$$\phi(x) \rightarrow \phi(x) - D\epsilon\phi(x) - \epsilon x^\mu \partial_\mu \phi(x). \quad (6.3)$$

The Lagrangian, on the other hand, must transform as

$$\mathcal{L} \rightarrow (1 + \epsilon)^{-4} \mathcal{L} = \mathcal{L} - 4\epsilon\mathcal{L} - \epsilon x^\mu \partial_\mu \mathcal{L} = \mathcal{L} - \epsilon \partial_\mu (x^\mu \mathcal{L}). \quad (6.4)$$

Therefore, the conserved current

$$D^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi + x^\mu \mathcal{L} = -\partial^\mu \phi (D\phi + x^\nu \partial_\nu \phi) + x^\mu \mathcal{L}. \quad (6.5)$$

Recall the energy-momentum tensor,

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}. \quad (6.6)$$

The conserved current becomes

$$D^\mu = x_\nu T^{\mu\nu} - \phi \partial^\mu \phi. \quad (6.7)$$